Reinforcement Learning I Introduction

Asad Jeewa



Housekeeping

- Available during my consultation hours
  - Fridays until 12pm for prayers
  - If unavailable for whatever reason, please send me an email.
- Lecture attendance is essential.
  - Engage fully with the module content as designed.
  - Ask questions in class.
  - Be proactive — don’t wait until test time to start learning.

Before we start this journey

- RL involves equations, proofs, and recursion.
- Aim to understand and develop intuition, not just memorise.
- The best way to learn is by implementing and coding it yourself.
- Take your time: the RL world unfolds step by step, so enjoy yourself while you learn and explore.

Introduction to Reinforcement Learning

Textbook

- Reinforcement Learning: An Introduction (R. Sutton & A. Barto)

Optional Extras

- Introduction to Deep Reinforcement Learning - Hugging Face Deep RL Course
- Tutorial: Reinforcement Learning; The Intuition Behind Policy Gradients by Dr. Arnu Pretorius
- Many more

What will be covered

- How does RL differ from ML?
- What is RL?
- Fundamental RL Concepts
- When is RL used?

The Big Picture

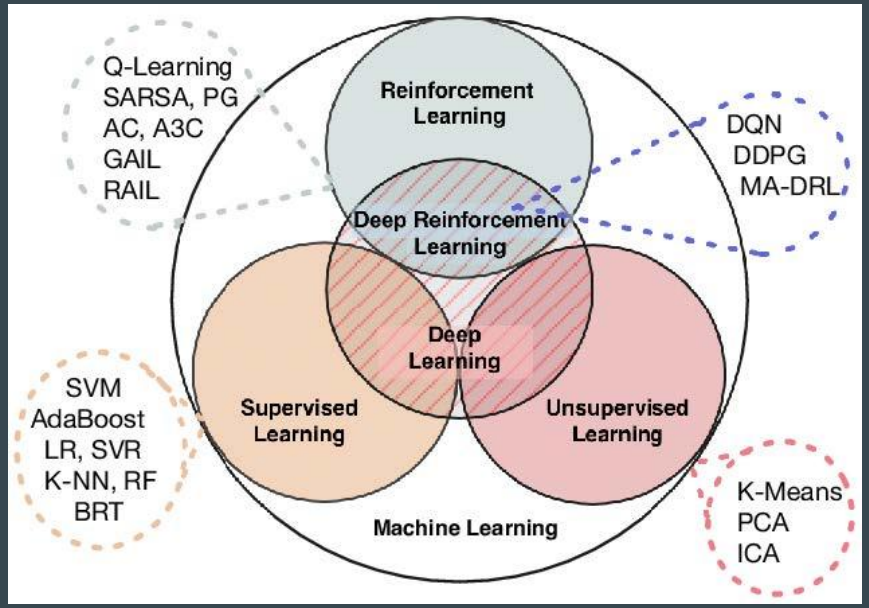
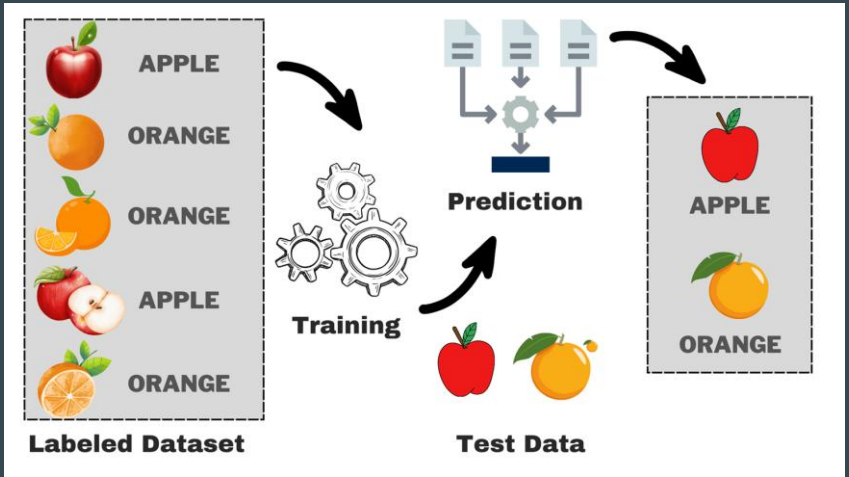
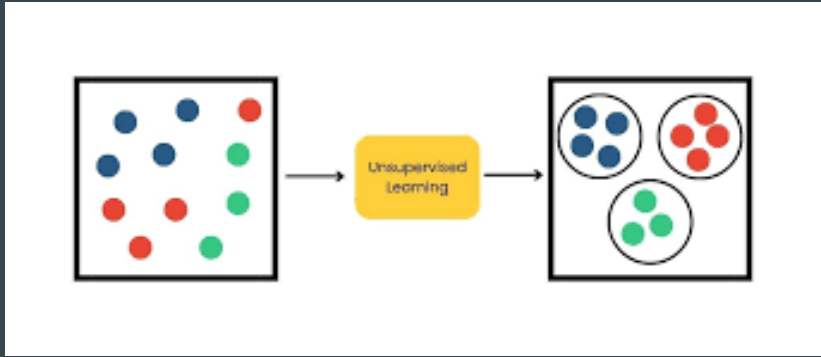


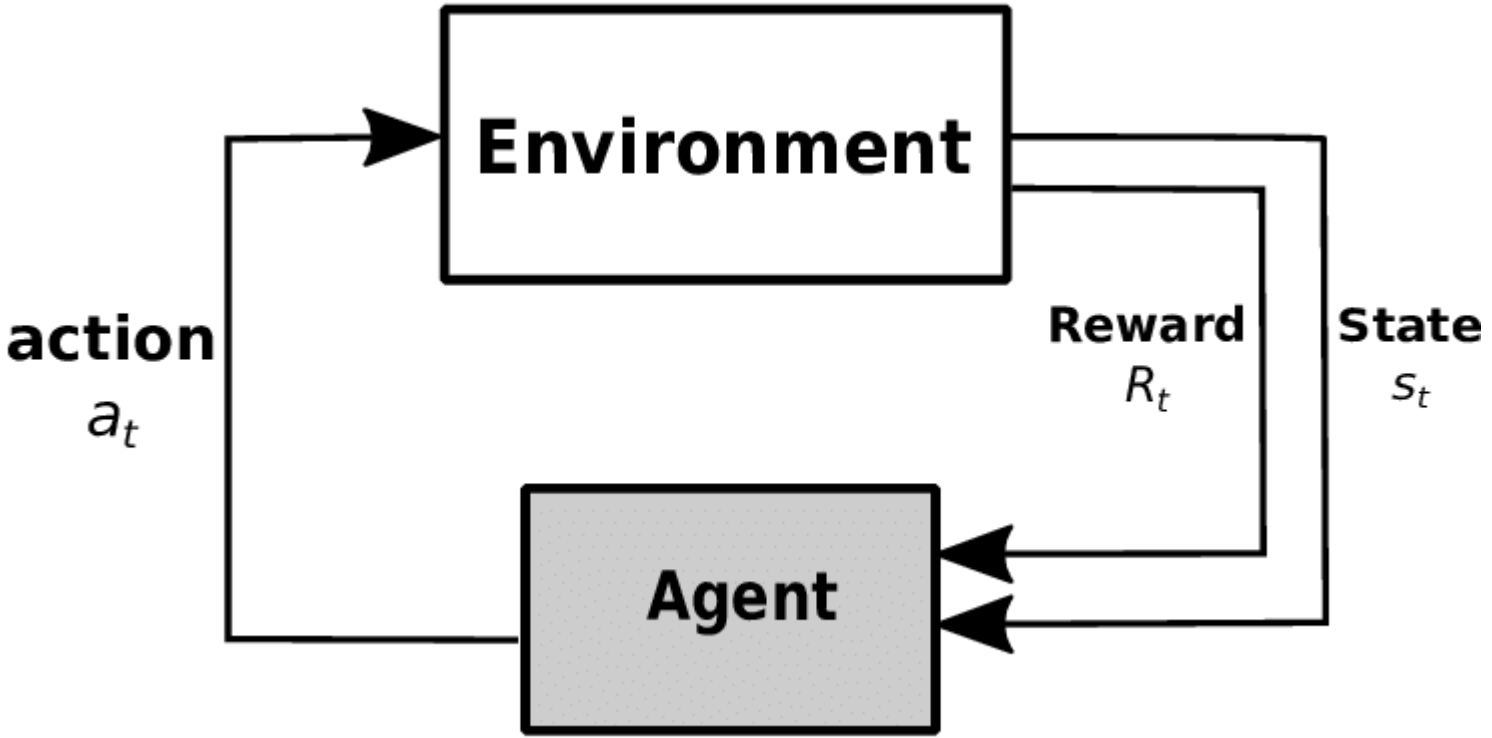
Figure: Taxonomy of Machine Learning [1]

Supervised Learning vs Unsupervised Learning





RL

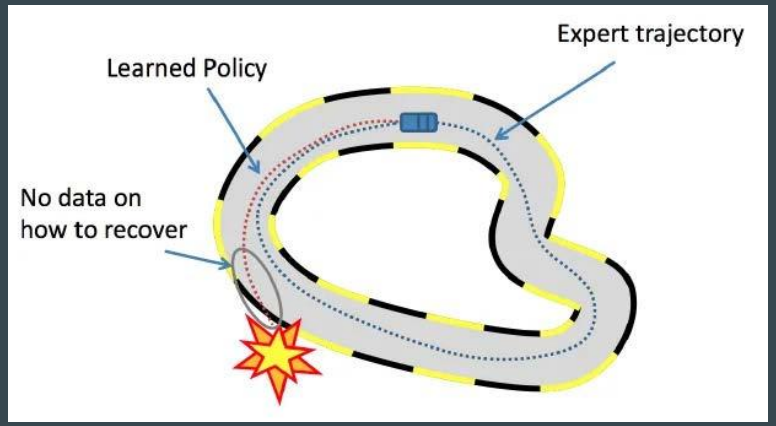


Relationship and difference between ML/RL, (un)supervised learning

- Machine Learning is a broad subfield of AI.
  - Develops algorithms and models that learn from and make decisions based on data, without being explicitly programmed for every possible scenario.
- 3 Learning paradigms under the *umbrella* of ML:
  - Supervised Learning: Learn from input-output pairs.
  - Unsupervised Learning: Learn patterns from unlabelled data.
  - Reinforcement Learning: :earn through trial and error from interaction with the environment

What about Imitation Learning?

- What about a supervised learning approach: predict actions from states?
- This approach is limited
- Cannot learn new behaviours
- Limited to the distribution of behaviour in the dataset



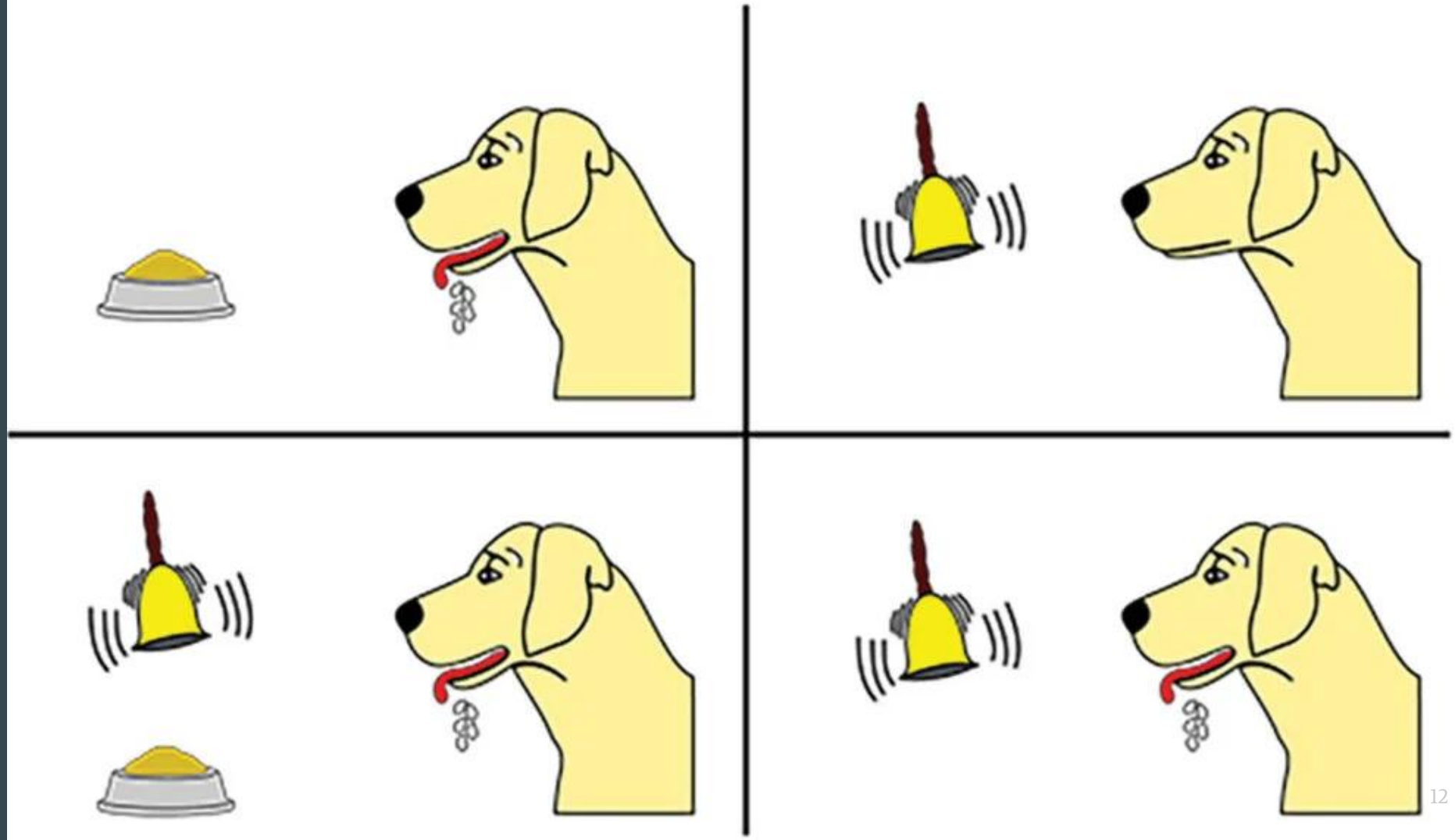
https://web.stanford.edu/class/cs237b/pdfs/slide/lecture_11_2024.pdf

Reinforcement Learning in a Nutshell

- Behaviour Learning
- Sequential Decision Making
- What do I do now to solve a problem later?
- How do my actions affect the environment?
- Powerful paradigm that has huge potential for real-world impact

"Imagine playing a new game whose rules you don't know; after a hundred or so moves, your opponent announces, 'You lose'."

- *Russell and Norvig (Introduction to AI)*



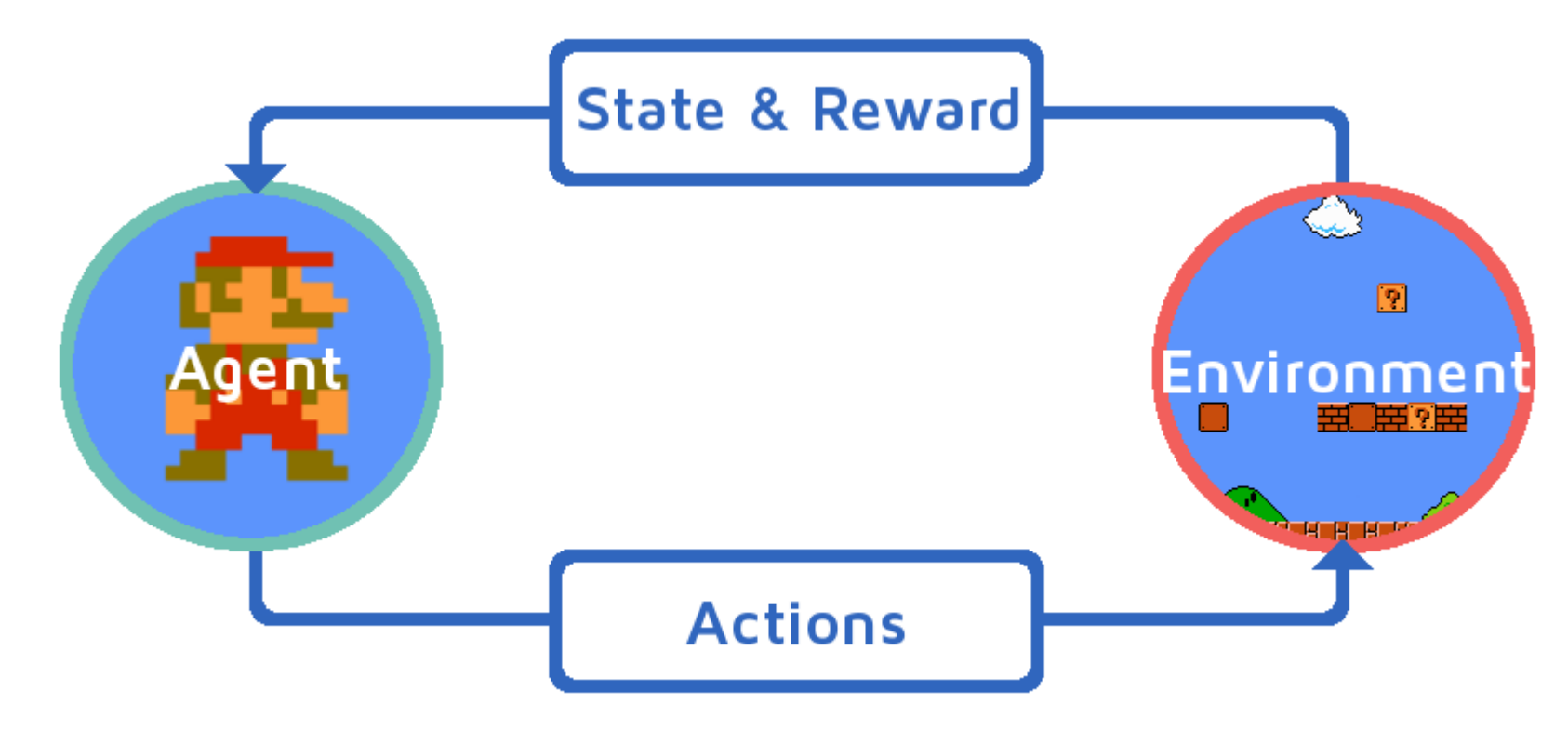
What is Reinforcement Learning (RL)?

What is Reinforcement Learning (RL)?

- Behaviour Learning: Learning to map states to actions.
- Sequential Decision Making: Multi-stage decision-making process.
  - What do I do now to solve a problem later?
  - How do my actions affect the environment?
- RL agent learns by trial and error.
- Agent actions may affect not only the immediate reward but also the future rewards.
- Powerful paradigm that has huge potential for real-world impact

What are the components of an RL Framework?

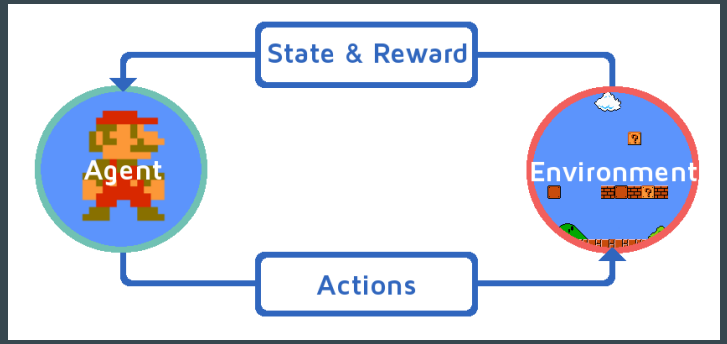
Components of an RL Framework



RL Loop

- Agent exists in an environment
- Agent takes actions based on the current state (view) of the
- environment
- Environment state updates
- Agent receives feedback as rewards (from the environment)

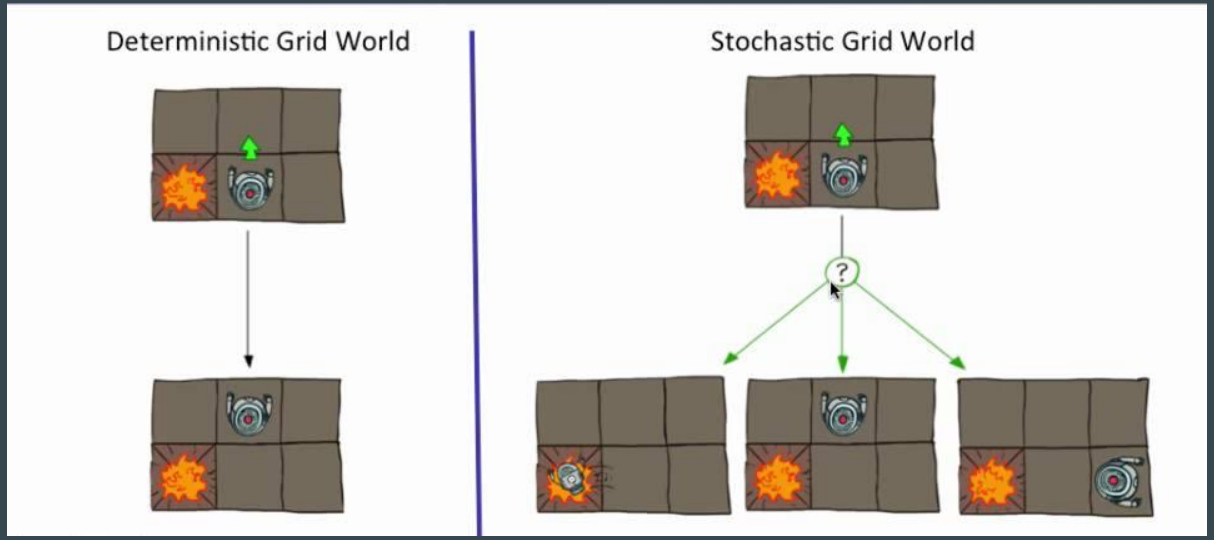
Learn to behave through trial and error without explicitly encoding the rules of the task



Fundamental RL Components

- Policy (States and Actions)
- Reward function (Value function)
- Environment

Representing Actions



Policy

- Policy π represents the agent's behaviour
- Mapping from states to actions
  - Deterministic: $a = \pi(s)$
  - Stochastic: $\pi(a|s) = P[a_t = a|s_t = s]$
- Goal of RL is to find an optimal policy π that maximises the expected discounted return

$$J(\pi) = E[\sum_{t=1}^{\infty}(\gamma^t r(s_t, a_t)]$$

Reward Function

The goal is to associate good behaviour with high rewards and bad behaviour with low rewards/ penalties

- Scalar feedback signal $R(s, a) = c$
- Sparse
- Delayed
- Reward engineering is more difficult than it seems!

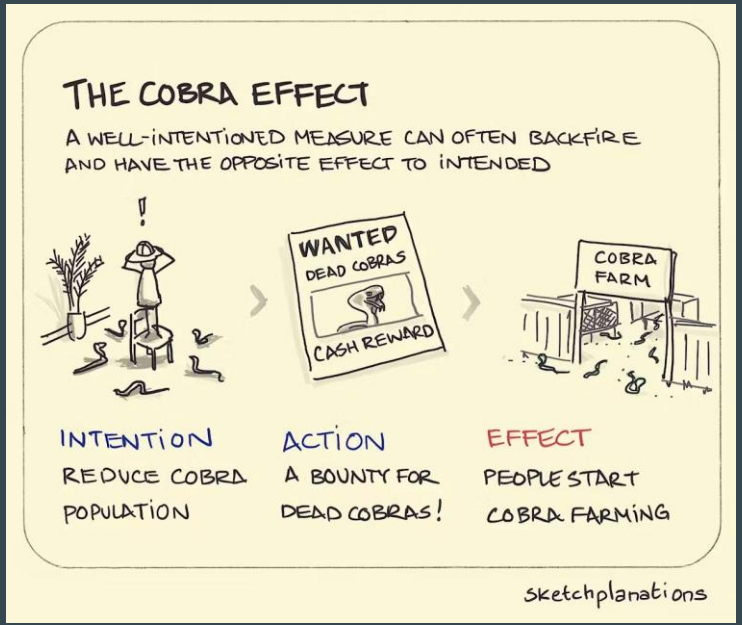
Reward Function





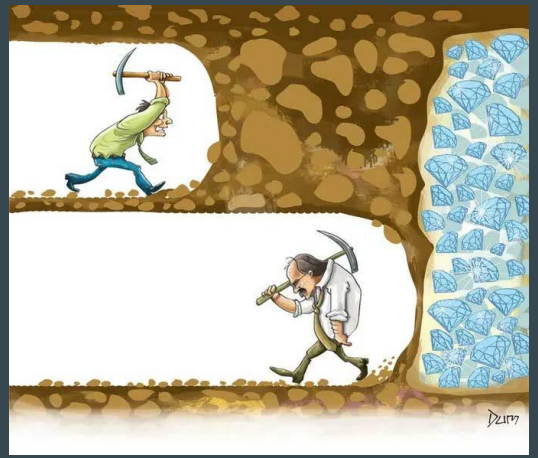


Rewards


THE COBRA EFFECT
A WELL-INTENTIONED MEASURE CAN OFTEN BACKFIRE AND HAVE THE OPPOSITE EFFECT TO INTENDED
!
WANTED
DEAD COBRAS
CASH REWARD
COBRA FARM
INTENTION
REDUCE COBRA POPULATION
ACTION
A BOUNTY FOR DEAD COBRAS!
EFFECT
PEOPLE START COBRA FARMING
sketchplanations

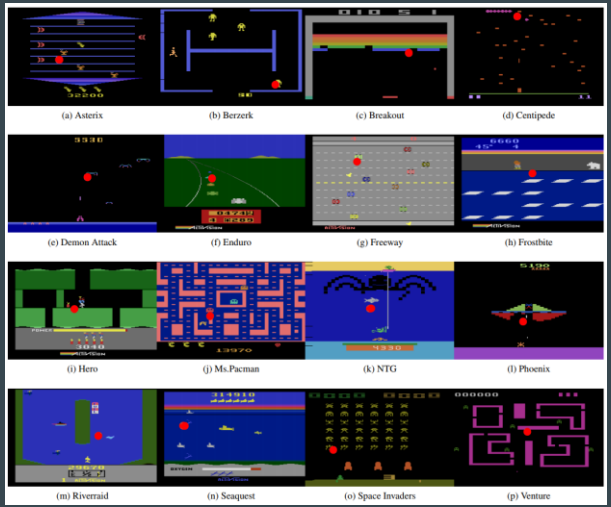
Delayed Rewards

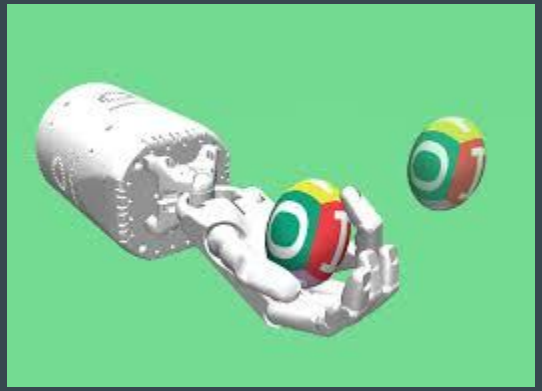
- Cannot rely on the instantaneous reward function
- Agent should not act myopically (short-term thinking)
- Need a notion of value that encodes the “goodness of a state”
- Goal is to associate actions in the present with the rewards that they influence in the future.
- Credit Assignment

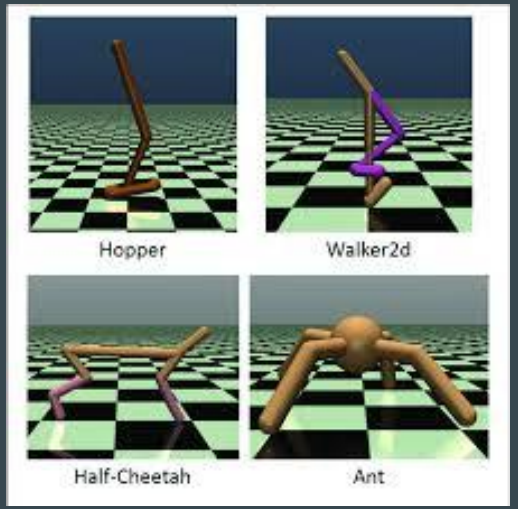


Environment

- The system or world the agent interacts with.
- Provides observations s_t to the agent and receives actions a_t in return.
- Returns a reward $r_t = r(s_t, a_t)$ signaling the success or failure of the action.
- Defines the dynamics $T(s_{t+1} \mid s_t, a_t)$ i.e. how the world changes.

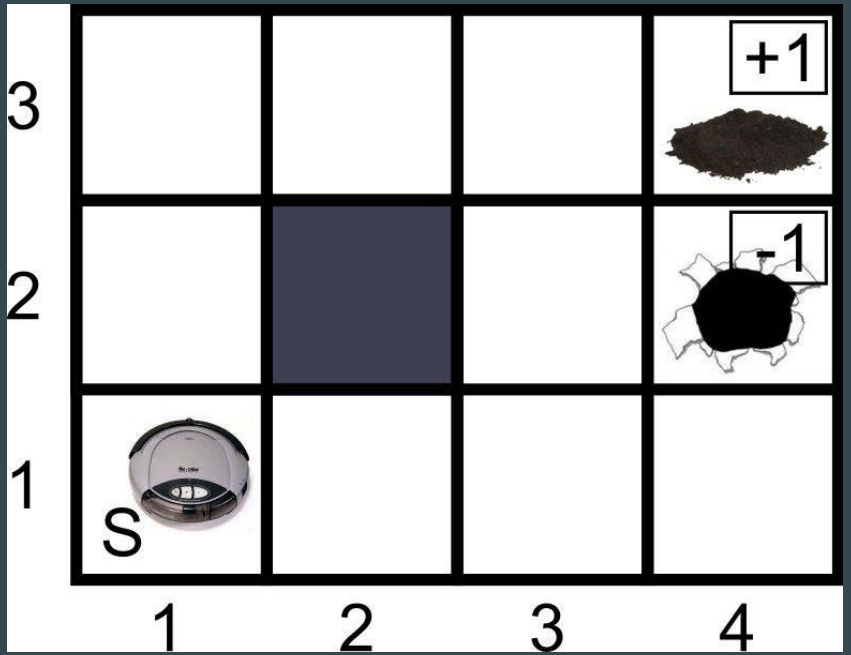

(a) Asterix
(b) Berzerk
(c) Breakout
(d) Centipede
(e) Demon Attack
(f) Enduro
(g) Freeway
(h) Frostbite
(i) Hero
(j) Ms.Pacman
(k) NTG
(l) Phoenix
(m) Riverraid
(n) Seaquest
(o) Space Invaders
(p) Venture




Hopper
Walker2d
Half-Cheetah
Ant



Example: Gridworld



- States: Position on grid
  - S is (1,1), goal (4,3)
  - OR grid of values
- Actions: {Up, Down, Left, Right}
- Reward:
  - +1 for finding dirt
  - -1 for falling into hole
  - -0.001 for every move
  - Why those specific values?

Reward Design Tips

- Rewards define desired behaviour: relative scale matters, not absolute magnitude
- Scaling all rewards by a constant does not change the optimal policy
- With neural networks, large rewards often means unstable training
  - It is wise to keep rewards small (e.g., $\sim[-1, 1]$) when possible.
  - Normalize, scale, or clip rewards only if needed for stability

RL in the real world

Examples of domains:

- Robotics: Teaching robots to grasp, walk, or manipulate objects.
- Autonomous vehicles: Adaptive navigation, traffic management, self-driving cars
- Finance: Portfolio optimisation and trading strategies.
- Healthcare: Treatment planning, personalised interventions.
- Video game playing
- ChatGPT!

- RL is widely applicable since we are learning behaviours.
- RL is ideal where explicit rules are hard to define.

Challenges

- High-dimensional state/action spaces.
- Delayed or sparse rewards.
- Safety and ethical constraints.
- Limited (on non-existent) online interaction with environment
- The world is unpredictable and messy!

Resources

- Sutton & Barto (1998), Reinforcement Learning: An Introduction.

Additional

- AlphaGo - The Movie | Full award-winning documentary

References

1. T. Yuan, W. Borba, K. Obraczka, C. Barakat, T. Turletti, C. Esteve Rothenberg, and W. da Rocha Neto, Harnessing Machine Learning for Next-Generation Intelligent Transportation Systems: A Survey. Sept. 2019.